Simplicial Sets, Quasicategories, and Type Theory

Tanner Altenkirk
Dalhousie University
Simplicial Type Theory Reading Group 2026

October 20, 2026

Overview

- The Simplex Category and Simplicial Sets
- Simplicial Relations and Lifting Properties
- Quasicategories and Kan Complexes
- Connections to Type Theory

The Simplex Category

Definition (The Simplex Category)

The simplex category $\mathbf{\Delta}$ is the category with:

- Objects: Finite, non-empty totally ordered sets $[n] := \{0, 1, 2, \cdots, n\}$.
- Morphisms: Order-preserving functions $f : [n] \longrightarrow [m]$.

Equivalently, $\mathbf{\Delta}$ is a full subcategory of **Cat** that has objects being posets defining finite non-empty ordinals.

Coface and Codegeneracy Maps

Definition (Coface Morphisms)

In $\mathbf{\Delta}$, there exist unique injective order-preserving functions $\delta^i : [n-1] \longrightarrow [n]$ for all $0 \leq i \leq n-1$ that skip the ith place in the image. We call these the *coface morphisms*.

Definition (Codegeneracy Morphisms)

In $\mathbf{\Delta}$, there exist unique surjective order-preserving functions $\sigma^i : [n+1] \longrightarrow [n]$ for all $0 \leq i \leq n+1$ that assign both i and $i+1$ in the domain to the ith place in the image. We call these the *codegeneracy morphisms*.

All morphisms in $\mathbf{\Delta}$ $f : [n] \longrightarrow [m]$ uniquely decompose into a sequence of coface and codegeneracy maps.

Simplicial Sets and Simplicial Objects

Definition (Simplicial Sets and Simplicial Objects)

A *simplicial set* is a contravariant functor from $\mathbf{\Delta}$ to **Set**. A *simplicial object* is a contravariant functor from $\mathbf{\Delta}$ to a category $\mathcal{C}$.

- We denote a simplicial set as $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Set}$.
- We denote the set of n-simplices of X as $X_n := X([n])$.
- The *category of simplicial sets* is denoted $\mathbf{sSet} := \mathbf{Set}^{\mathbf{\Delta}^{\mathrm{op}}} = \mathit{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathbf{Set})$.
- $X(\delta^i) := d_i$ is called a *face* of X.
- $X(\sigma^i) := s_i$ is called a *degeneracy* of X.
- Decomposition: $X(f) = s_{i_l} \cdots s_{i_0} d_{j_k} \cdots d_{j_0}$, $j_0 < \cdots < j_k$, $i_0 < \cdots < i_l$.

Simplicial Relations

The following are known as the *simplicial relations*:

- $d_i d_j = d_{j-1} d_i$ if $i < j$.
- $s_i s_j = s_{j+1} s_i$ is $i \leq j$.
- $$d_i s_j = \begin{cases} s_{j-1} d_i & \text{if } i < j \\ 1 & \text{if } i = j \text{ or } i = j+1 \\ s_j d_{i-1} & \text{if } i > j+1 \end{cases}$$

We can also define a simplicial object as a collection of objects $(X_n)_{n \in \mathbb{N}}$ with $d_i : X_{n+1} \longrightarrow X_n$ and $s_i : X_n \longrightarrow X_{n+1}$ for $0 \leq i \leq n$ satisfying the simplicial relations above. We denote the *standard n-simplex* in **sSet** as Δ^n, the simplex representing $[n]$. We then define

$$\Delta^n_k := \mathbf{\Delta}([k], [n])$$

A simplex $x \in X_n$ is called *degenerate* if $x = s_i(y)$ for some $y \in X_{n-1}$, $i \leq n-1 \in \mathbb{N}$.

Boundaries of Simplicial Sets

Definition (Simplicial Subset of a Simplicial Set)

A *simplicial subset* $T_n \subset X_n$ is a sequence of subsets of X_n such that for all $0 \leq i \leq n$, $d_i(T_n) \subset T_{n-1}$ and $s_i(T_n) \subset T_{n+1}$. This defines a simplicial set structure on T_n.

The image $d_i(\Delta^{n-1})$ in Δ^n is called the ith *face* of Δ^n.

Definition (The Boundary of a Simplicial Set)

the *boundary* of a simplicial set X is a sequence of $(n-1)$-simplices $x_i \in X_{n-1}$ (denoted $(x_0, \cdots, x_n)$) such that $d_i(x_j) = d_{j-1}(x_i)$ for $i < j \leq n$. the boundary of Δ^n is denoted

$$\partial\Delta^n_k := \{f : [k] \longrightarrow [n] | \text{ f not surjective}\}$$

We could also define the boundary of Δ^n as the union of the ith faces of Δ^n for all $0 \leq i \leq n$.

Horns

Definition (k^{th} Horn of Δ^n)

For $0 \leq k \leq n$, the kth *horn in* Δ^n is is the simplicial subset Λ^n_k defined as:

$$(\Lambda^n_k)_m := \{f : [m] \longrightarrow [n] | [n] \not\subset f([m]) \cup k\}$$

Thus, Λ^n_k is all the faces of Δ^n except the kth one.

Definition (General Horns)

A *horn* $\Lambda^n_k \longrightarrow X$ in a simplicial set X is a list $(x_0, \cdots, x_{k-1}, *, x_{k+1}, \cdots x_n)$ containing $\prod_{i \neq k} X_{n-1}$ such that $d_i(x_j) = d_{j-1}(x_i)$ for $i < j$ and $i, j \neq k$.

Note: A horn $\Lambda^k_n \longrightarrow X$ is *inner* if $0 < k < n$, and *outer* if $k = 0$ or $k = n$.
Big Question: When can we extend a horn $\Lambda^k_n \longrightarrow X$ to a simplicial set Δ^n?

Lifting Properties

Definition (Right Lifting Properties)

Let $i : A \longrightarrow B$ and $p : X \longrightarrow Y$. Then p has the *right lifting property* (RLP) with respect to i if for any two maps $a : A \longrightarrow X$ $b : B \longrightarrow Y$, we have a lift $h : B \longrightarrow X$ such that $h \circ i = a$ and $p \circ h = b$.

$$\begin{array}{ccc} A & \xrightarrow{a} & X \\ {\scriptstyle i}\downarrow & \overset{\exists h}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[b]{} & Y \end{array}$$

The *left lifting property* (LLP) is the same as the setup above, except now $a : X \longrightarrow A$ and $b : Y \longrightarrow B$ and the lift $h : Y \longrightarrow A$ satisfies $i \circ h = b$, $h \circ p = a$.

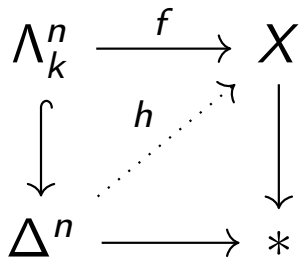
Quasicategories and Kan Complexes

Definition (Quasicategory)

A *quasicategory* is a simplicial set X such that the unique map $X \longrightarrow *$ has the RLP with respect to all inner horn inclusions. (These are $(\infty, 1)-$categories!)

Definition (Kan Complex)

A *Kan complex* is a simplicial set X such that the unique map $X \longrightarrow *$ has the RLP with respect to all horn inclusions. (These are $\infty-$groupoids!)



These lifts are sometimes called *horn fillings*.

Homotopy, Quasicategories and Kan Complexes

In a Quasicategory $\mathcal{C}$ with $x, y \in \mathcal{C}_0$ and $f : x \longrightarrow y$, $g : z \longrightarrow x$, and $h : w \longrightarrow z$ in $\mathcal{C}_1$:

- Composition $f \circ g$ is encoded by a 2-simplex on f and g.
- Identities are encoded by degeneracies $s_0 : \mathcal{C}_0 \longrightarrow \mathcal{C}_1$ taking x to $1_x : x \longrightarrow x$.
- Associativity is encoded by a 3-simplex expressing $(f \circ g) \circ g \simeq f \circ (g \circ h)$ (hence NOT strictly associative).
- Unitality is encoded by degeneracies expressing $1_y \circ f \simeq f$ and $f \circ 1_x \simeq f$.

A Kan complex behave similarly, but since all horns have fillings, composition of 1-simplices is also only defined "up to homotopy".

Model Category Theorem

Kan complexes and quasicategories are objects of so called *model categories*. There really is too much here to cover in great detail. For the purposes of this theorem:

- A monomorphism of simplicial sets is called a *cofibration*.
- A weak homotopy equivalence between the geometric realizations of simplicial sets is called a *weak equivalence*.
- A horn lift (sometimes called a *Kan fibration*) is called a *fibration*.
- A monomorphism of simplicial sets that also yeilds a weak homotopy equivalence of the geometric realizations of these simplicial sets (ie. both a cofibration and a weak equivalence) is called a *trivial cofibration*.

Theorem (A Theorem of Model-Categories)

In a model category, if i is a trivial cofibration and p is a fibration, then i has the LLP with respect to all fibrations, and p has the RLP with respect to all trivial cofibrations.

This will help us connect to (Homotopy/Simplicial) type theory.

Homotopy Type Theory and Kan Complexes

Simplicial Sets	**Homotopy Type Theory**
A a Kan complex	The type of $x : A$
A^{Δ^1} Kan complex of maps $\Delta^1 \longrightarrow A$	The type of a path p
$r : A \longrightarrow A^{\Delta^1}$ constant map	Assignment $x \mapsto \mathrm{refl}_x$
$\chi : P \longrightarrow A^{\Delta^1}$ fibration of family P	Dependent type $P(x, y, p)$
$d : A \longrightarrow P$ with $\chi \circ d = r$	Term of constant path $P(x, x, \mathrm{refl})$
Lift s of with $\chi \circ s = \mathrm{id}_{A^{\Delta^1}}$, s$\circ r = d$	Path Induction!

The lift s follows from the theorem of model categories, because χ is a fibration and r is a trivial cofibration.

Homotopy Type Theory and Kan Complexes cont.

Simplicial Sets	**Homotopy Type Theory**
A a Kan complex	The type of $x : A$
A^{Δ^1} Kan complex of maps $\Delta^1 \longrightarrow A$	The type of a path p
$r : A \longrightarrow A^{\Delta^1}$ constant map	Assignment $x \mapsto \mathrm{refl}_x$
$\chi : P \longrightarrow A^{\Delta^1}$ fibration of family P	Dependent type $P(x, y, p)$
$d : A \longrightarrow P$ with $\chi \circ d = r$	Term of constant path $P(x, x, \mathrm{refl})$
Lift s of with $\chi \circ s = \mathrm{id}_{A^{\Delta^1}}$, $s \circ r = d$	Path Induction!

